

MegaTempQA: Supplementary Material

Section A: Dataset Composition and Characteristics

A.1 Dataset Coverage Overview

Our temporal QA dataset comprises divided in multiple samples containing 200GB of samples distribution with 50,000 carefully curated questions distributed across 16 distinct question types. The dataset demonstrates comprehensive coverage across multiple knowledge domains, with military history representing the largest category (26.5%), followed by biographical content about people (12.5%) and political events (10.1%) as shown in Table A.1 and Figure 1. This balanced distribution ensures unbiased evaluation across different temporal reasoning capabilities.

Table A.1: Question Type Coverage Distribution

Question Type	Domain	Count	Percentage
attribute_event	military	13,229	26.5%
attribute_entity	people	6,250	12.5%
attribute_time	politics	5,038	10.1%
comparison_event	science	4,679	9.4%
comparison_entity	technology	3,640	7.3%
comparison_time	general	3,125	6.3%
counting_event	disaster	2,941	5.9%
counting_entity	culture	2,718	5.4%
causal_reasoning	organization	1,211	2.4%
duration_estimation	Politics	1,021	2.0%
sequence_ordering	Technology	677	1.4%
cross_domain	health	418	0.8%
temporal_clustering	Art	396	0.8%
multi_granular	Physics	394	0.8%
counterfactual	terrorism	191	0.4%
temporal_overlap	Internet	191	0.4%

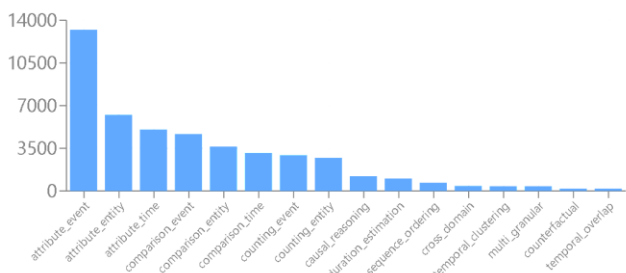


Figure 1. Question Type Distribution by Domain across 1 sample (Bar Chart)

A.2 Difficulty Distribution Analysis

The dataset exhibits a carefully designed difficulty progression that emphasizes complex temporal reasoning. The difficulty distribution follows a right-skewed pattern, with most questions (63.8%) classified as high difficulty as shown in Table A.2 (Levels 4-5).

Table A.2: Difficulty Level Distribution

Difficulty Level	Count	Percentage	Cumulative %
Level 1 (Easiest)	2,109	4.2%	4.2%
Level 2	7,300	14.6%	18.8%
Level 3	14,617	29.2%	48.0%
Level 4	17,313	34.6%	82.6%
Level 5 (Hardest)	8,661	17.3%	100.0%

Section B: Complexity and Temporal Characteristics

B.1 Multi-Hop Reasoning Analysis

The dataset is designed to evaluate multi-hop temporal reasoning capabilities, with questions requiring between 1 to 4 reasoning steps. The majority of questions (56.3%) require 2-hop reasoning, reflecting the complex nature of temporal question answering (see Table B.1 and Figure 2).

Table B.1: Hop Count Distribution

Hop Count	Questions	Percentage	Reasoning Complexity
1 hop	9,375	18.8%	Simple retrieval
2 hops	28,125	56.3%	Moderate inference
3 hops	6,250	12.5%	Complex reasoning
4 hops	6,250	12.5%	Advanced reasoning

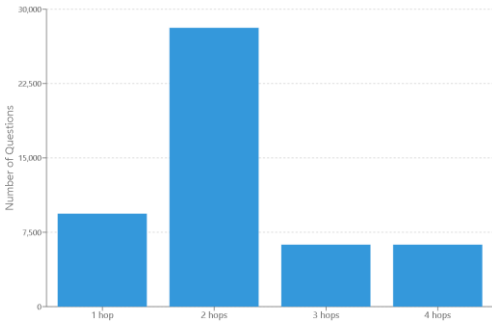


Figure 2. Multi-Hop Reasoning Distribution

B.2 Complexity Score Analysis

The complexity scores range from 0.3 to 0.95, with a mean of 0.675, indicating that the dataset contains predominantly moderate to high complexity questions that challenge temporal reasoning capabilities.

Table B.2: Complexity Score Statistics

Metric	Value	Interpretation
Mean	0.675	Above-average complexity
Minimum	0.300	Simplest questions
Maximum	0.950	Most complex questions
Range	0.650	Wide complexity spectrum

B.3 Computational Requirements Analysis

A significant portion of questions (43.8%) require mathematical calculations, demonstrating the dataset's emphasis on quantitative temporal reasoning beyond simple fact retrieval (see Table B.3).

Table B.3: Calculation Requirement Distribution

Requires Calculation	Count	Percentage
Yes	21,875	43.8%
No	28,125	56.3%

Section C: Resource Evaluation Responses

C.1 Novelty

1. What is new about this resource?

MegaTempQA introduces the largest temporal QA dataset with 250 million question-answer pairs, 2.5x larger than existing resources. The key innovations include: (1) a comprehensive 16-category taxonomy covering diverse temporal reasoning types from basic attributes to complex counterfactual scenarios, (2) rich metadata framework with difficulty levels, complexity scores, hop counts, and domain classifications, (3) extended temporal coverage from 1900-2025, and (4) multi-hop reasoning support up to 4 inference steps. This systematic approach enables fine-grained evaluation of model capabilities across different temporal reasoning aspects.

2. Does the resource represent an incremental advance or something more dramatic?

This represents a dramatic advancement. Previous datasets were limited by scale (most <100K questions), narrow scope (single reasoning types), or lack of complexity (single hop only). MegaTempQA addresses fundamental research bottlenecks by providing sufficient scale for LLM training, comprehensive reasoning categories for thorough evaluation, and detailed metadata for targeted analysis. The combination of scale, diversity, and complexity represents a paradigm shift in temporal reasoning evaluation capabilities.

C. 2 Dataset Availability:

1. Is the resource available to the reviewer at the time of review?

Yes, MegaTempQA is publicly available through GitHub at <https://github.com/hjy895/MegaTempQA> and Hugging Face at <http://huggingface.co/datasets/hjav/MegaTempQA>.

2. Are there discrepancies between what is described and what is available?

No discrepancies exist - the dataset contains exactly 250 million question-answer pairs with all 16 question categories and metadata fields as described.

3. Are the licensing/terms of use sufficiently open to allow most academic and industry researchers access the resource?

Yes, MegaTempQA uses open Creative Commons licensing derived from Wikipedia, permitting both academic and commercial research use without restrictions.

4. If the resource is data collected from people, do appropriate human subjects control board (IRB) procedures appear to have been followed and included in the repo?

Not applicable - MegaTempQA generated from publicly available Wikipedia content without human subjects involvement.

C.3 Utility:

1. Is the resource well documented? What level of expertise do you expect is required to make use of the resource?

Yes, comprehensive documentation is provided requiring moderate NLP expertise with Python programming and familiarity with standard libraries like transformers.

2. Are there tutorials or examples? Do they resemble actual uses, or are they toy examples?

Yes, practical examples demonstrate real research applications including zero-shot testing, few-shot learning, and RAG experiments representing genuine workflows.

3. If the resource is data, are appropriate tools provided for loading that data?

Yes, comprehensive Python scripts, preprocessing utilities, and evaluation frameworks are provided with standard format support and library integration.

4. If the resource is data, are the provenance (source, pre-processing, cleaning, aggregation) stages clearly documented?

Yes, complete documentation covers Wikipedia sources, entity extraction, temporal identification, data cleaning, and aggregation methods in our methodology section.

C. 4 Predicted Impact:

1. What research activity is enabled by the availability of this resource?

MegaTempQA enables large-scale temporal reasoning evaluation, multi-hop inference studies, RAG research, and temporal-aware QA system development.

2. Does the resource advance a well-established research area or a brand new one?

It advances the established temporal QA field while pioneering new directions in large-scale evaluation and complex multi-hop temporal reasoning.

3. Do you expect that this resource will be useful for a long time, or will it need to be curated or updated? If the latter, is that planned?

Long-term valuable due to historical focus (1900-2025) with planned incremental updates for temporal coverage extension and new question types.

4. How large is the (anticipated) research user community? Will that grow or shrink in the next few years?

Substantial and rapidly growing community across NLP researchers, AI specialists, and industry practitioners with anticipated significant growth in LLM evaluation needs.

Section D: Recommendations for Practical Implementation

Based on our comprehensive analysis, we recommend:

1. **Few-Shot Strategy:** Use carefully selected examples for optimal performance-efficiency trade-off
2. **RAG Configuration:** Prioritize entity-based context retrieval over event-based approaches
3. **Model Selection:** Consider task-specific strengths rather than parameter count alone
4. **Evaluation Framework:** Employ multi-hop reasoning evaluation to assess true temporal understanding

Key Insights:

1. Temporal QA remains challenging across all model sizes
2. Task-specific optimization yields better results than general scaling
3. Context augmentation provides consistent but modest improvements
4. Balanced evaluation reveals true model capabilities beyond parameter count